SNAP Participation and Household Calorie Intake: Evidence from FoodAPS

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1 Introduction

The Supplemental Nutrition Assistance Program (SNAP) serves over 40 million Americans annually, yet its causal effect on calorie intake remains contested. Simple comparisons are misleading because SNAP recipients differ from non-recipients on income, household size, and other determinants of food consumption. This paper estimates the average treatment effect on the treated (ATT) using five estimators applied to the USDA FoodAPS data: OLS, nearest-neighbour matching (NN), propensity score matching (PSM), inverse probability weighting (IPW), and doubly-robust IPW regression adjustment (IPWRA).

2 Data and Sample

I merge the FoodAPS nutrient file with the household characteristics file, collapse to the household level, and winsorise outcomes at the 1st–99th percentile to limit the influence of extreme values (raw maximum: 49,278 kcal). The analytic sample contains **4,303 households**: 1,342 SNAP recipients (31.2%) and 2,961 non-recipients (68.8%).

The primary outcome is **winsorised calories per capita** (`calories_pc_w`). Total calories is a secondary outcome; it confounds programme effects with household size, since SNAP households are on average larger ($\bar{n} = 3.5$ vs. $\bar{n} = 2.4$ members). SNAP recipients consume fewer calories *per capita* (72.7 vs. 86.7 kcal) despite higher household totals, motivating the per-capita focus.

3 Empirical Strategy

3.1 Estimating Equation

$$Y_i = \beta_0 + \beta_1 \, \text{SNAP}_i + X_i' \gamma + \varepsilon_i \tag{1}$$

Controls X_i include income, household size, rurality, census region fixed effects, food sufficiency, liquid assets, car access, store distance, and perceived cost and time barriers to healthy eating.

3.2 Identification Assumptions

All five estimators rely on three conditions:

1. **Unconfoundedness (ignorability):** $(Y^0, Y^1) \perp D \mid X$. SNAP assignment is as good as random conditional on X. This is plausible given the rich covariate set but cannot be verified; unobserved factors (dietary preferences, health literacy) may still confound the estimate.

2. **Overlap:** $0 < \Pr(D = 1 \mid X) < 1$. Figures 1–2 show substantial common support; trimming removes only 33 observations (0.8%).

3. **SUTVA:** No interference between households and a single version of SNAP treatment. Plausible given the household-level unit of analysis.

3.3 Doubly-Robust IPWRA

IPWRA is consistent if *either* the outcome model or the propensity score model is correctly specified—not necessarily both. This double-robustness property makes it the preferred estimator when model specification is uncertain [1].

3.4 Covariate Balance and Robustness

The propensity score (logit, pseudo-$R^2 = 0.222$) is dominated by income and household size. Two covariates—`healthycost` and `healthytime`—exhibit pathological variance ratios ($>$ 3,000) after matching, indicating near-degenerate distributions. I conduct three robustness checks: (i) trimming to the common support interval, (ii) caliper matching at $0.20\,\hat{\sigma}_{\hat{p}} = 0.046$, and (iii) PSM excluding the two problematic covariates. The caliper was non-binding (all pairs satisfied it), while the clean PSM—which achieves full balance ($|\text{std. diff.}| < 0.10$ on

all covariates)—produces an ATT of -420 kcal ($p = 0.103$), substantively larger than the base PSM.

4 Results

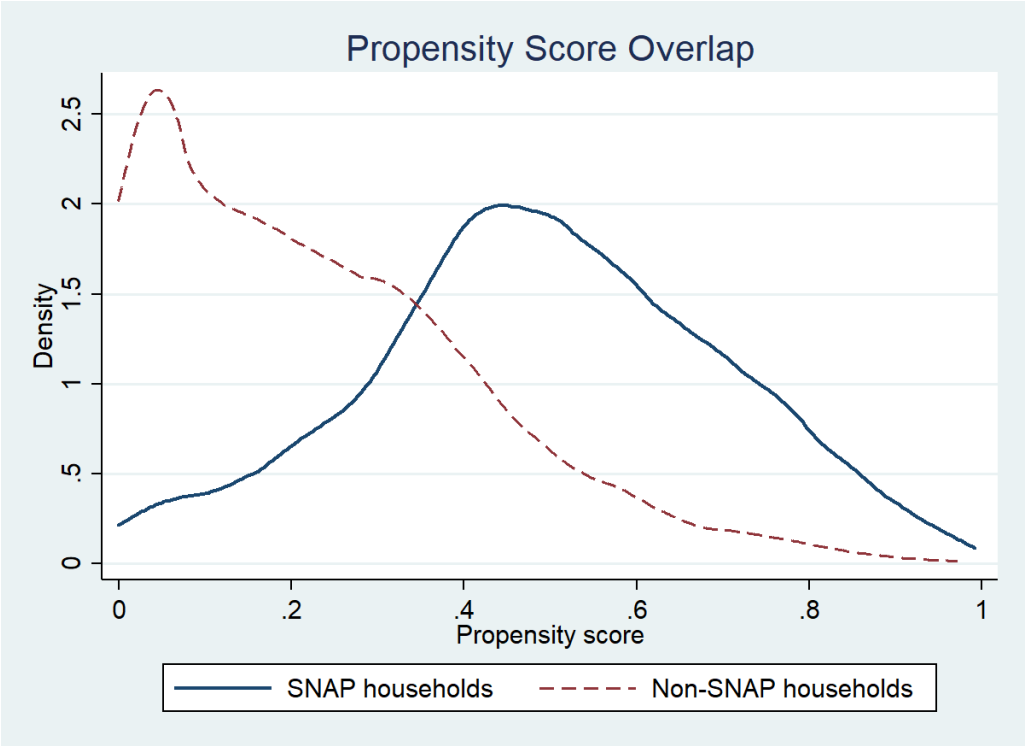

Figure 1: **Propensity Score Overlap.** SNAP households (solid) vs. non-SNAP (dashed). Substantial overlap exists in [0.2, 0.9], but a mass of non-SNAP households concentrates near $\hat{p} \approx 0$, creating a density gap rather than a gap in support.

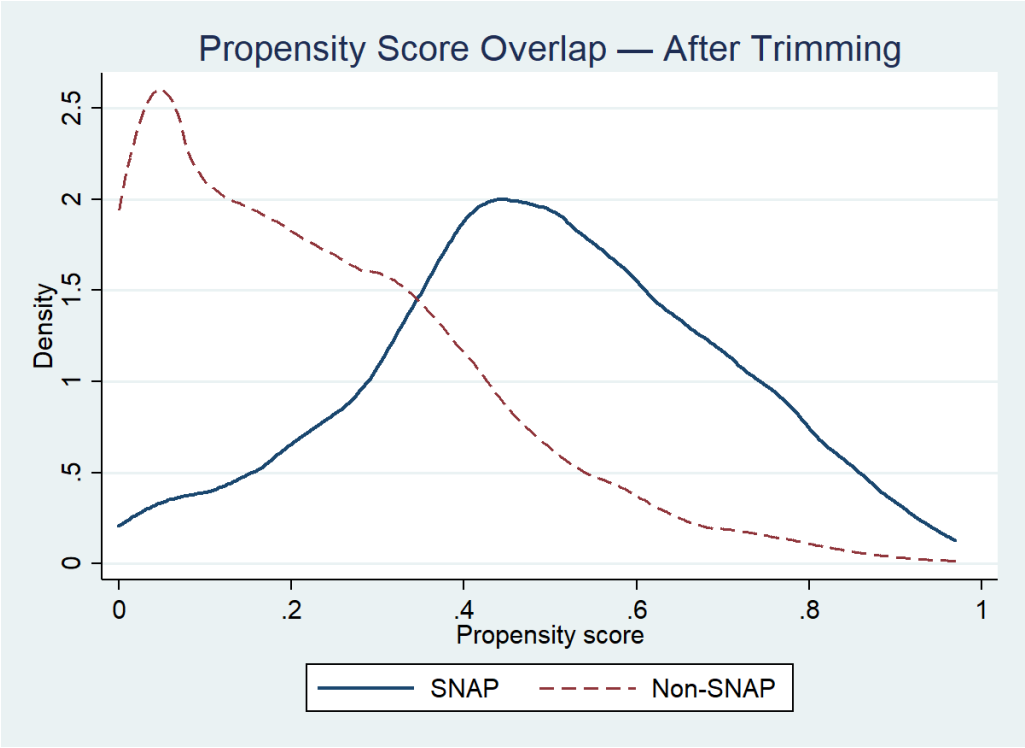

Figure 2: **Propensity Score Overlap After Trimming.** Removing 33 tail observations (0.8%) yields near-identical distributions, confirming the overlap problem is confined to a small extreme tail.

Table 1 presents ATT estimates for the primary outcome (winsorised calories per capita) across all five estimators.

Table 1: Effect of SNAP on Calories Per Capita (Winsorised, ATT)

	OLS	**NN**	**PSM**	**IPW**	**IPWRA**
SNAP participation	−1.84	−0.21	3.35	4.28**	−3.82**
	(1.48)	(1.76)	(2.06)	(2.06)	(1.59)
Observations	4,302	4,302	4,302	4,302	4,302
R^2	0.510	—	—	—	—

Robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.
Outcome winsorised at 1st–99th percentile. Controls: income, household size, rurality, food sufficiency, liquid assets, car access, store distance, healthy-food cost and time, census region FE.
IPWRA = IPW Regression Adjustment (doubly robust).

OLS and matching estimators. OLS gives −1.84 kcal/person ($p = 0.215$); NN and PSM give −0.21 and +3.35 kcal respectively, all insignificant. The magnitude of the OLS

estimate is economically trivial—less than 2.5% of the mean non-SNAP per-capita calorie intake of 76.3 kcal—and its sign is sensitive to covariate specification.

IPW and IPWRA: sign divergence and model dependence. IPW gives +4.28 kcal ($p = 0.038$) while IPWRA gives −3.82 kcal ($p = 0.016$)—opposite signs, both statistically significant. This divergence is a classic symptom of **model dependence**: the result is sensitive to whether the outcome regression or only the propensity score enters the estimator. IPW uses only the propensity score; IPWRA additionally fits separate outcome regressions for treated and control groups, which better accounts for the nonlinear relationship between household size and per-capita calories. Because IPWRA is consistent under a strictly weaker condition—misspecification of either but not both models is tolerable—it is the preferred estimate.

Preferred estimate. The doubly-robust IPWRA estimate of −3.82 kcal per person per household ($p = 0.016$, 95% CI: [−6.94, −0.70]) is the most credible point estimate. Its magnitude is modest—roughly 5% of the counterfactual mean of 76.3 kcal—consistent with SNAP *maintaining* rather than substantially increasing per-capita calorie adequacy for participating households.

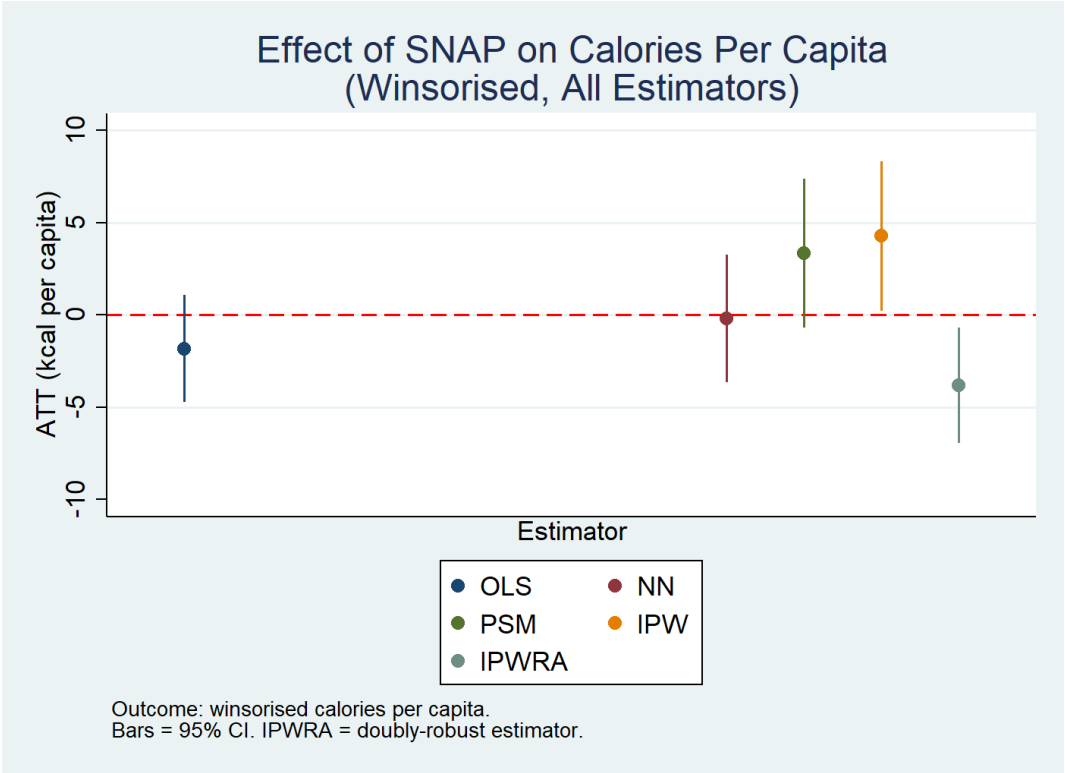

Figure 3: **ATT Estimates Across All Five Estimators.** Each point is the estimated effect of SNAP on winsorised per-capita calories; bars show 95% CI. IPW ($+4.3$, $p = 0.038$) and IPWRA (-3.8, $p = 0.016$) are statistically significant but have opposite signs, illustrating model dependence. IPWRA (doubly-robust) is the preferred estimate.

5 Discussion and Conclusion

The sign divergence between IPW and IPWRA is not a failure of the analysis—it is informative. It reveals that the result is sensitive to functional-form assumptions about the outcome regression, which is precisely the situation where IPWRA's double-robustness property matters most. Under IPWRA, SNAP participation is associated with a small but statistically significant *reduction* in per-capita calorie intake of approximately 3.8 kcal/person. This raises concerns about **selection on unobservables**: households that select into SNAP may have lower underlying caloric need or different dietary preferences that persist even after observable controls. Matching on observables cannot rule this out, and a design exploiting exogenous eligibility variation would be needed for sharper identification.

The inconsistency across estimators also highlights a methodological lesson: reporting a single causal estimator is insufficient when model dependence is present. The appropriate response is to report the full range of estimates, explain why they differ, and identify the

most credible one on theoretical grounds—which is what this paper does.

AI Use Statement

Claude (Anthropic) assisted with Stata pipeline structure, debugging, and LaTeX formatting. All analytical decisions, variable construction, identification arguments, and interpretations are my own.

References

[1] Bang, H., and Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4), 962–972.

[2] USDA Economic Research Service (2016). *National Household Food Acquisition and Purchase Survey (FoodAPS) Public Use File*. Washington, DC: USDA ERS.

[3] Rosenbaum, P. R., and Rubin, D. B. (1985). Constructing a control group using multivariate matched sampling methods that incorporate the propensity score. *The American Statistician*, 39(1), 33–38.